

Type 1 vs Type 2 virtualization

- **A Type 1(Bare-Metal Virtualization) hypervisor runs directly on the host's hardware, without needing an underlying operating system.**
- **This type provides direct access to hardware resources, making it more efficient and better suited for enterprise environments that require high performance, scalability, and stability.**
- **VMware ESXi, Microsoft Hyper-V, Xen Hypervisor, KVM**
- **A Type 2(Hosted Virtualization) hypervisor runs on top of a host operating system (OS) rather than directly on the hardware.**
- **It depends on the host OS to handle basic system resources and hardware interactions, which makes it easier to set up and use on personal devices.**
- **VMware Workstation, Oracle VirtualBox, Parallels Desktop, QEMU**

Feature	Type 1 (Bare-Metal)	Type 2 (Hosted)
Performance	High performance with direct hardware access	Lower performance, depends on host OS
Setup Complexity	More complex, often used in data centers	Simple setup, ideal for personal use
Use Cases	Cloud providers, enterprise data centers, HPC	Development, testing, education
Examples	VMware ESXi, Microsoft Hyper-V, Xen, KVM	VMware Workstation, VirtualBox, Parallels
Ideal Environment	Production servers with dedicated hardware	Personal desktops and development setups